

Costing of producing DPS

This report will explain the estimated cost for manufacture of our boards, including the component, materials, design and test costs, etc. It will show the assumption of one board and the massive production (200 units) with marginal cost. The unit cost is made up of recurring cost and non-recurring cost per unit.

Recurring cost:

The total recurring cost for a single board is , and for one unit from 200 boards is.

- **Materials**

The following table illustrates the component lists for each board. There is a comparison between the cost of one board and the unit price for quantities used of 200 boards. Although some companies may offer a lower price, they ask for a higher delivery fee or only provide large quantities. We try to find the most appropriate suppliers and list the websites in the reference.

Component	Part number	Quantity/unit	Cost for 1 board	Cost for 200 boards
0805Resistor	1	2	0.0208*2	£0.006*2
0805Resistor	24	2	0.0208*2	£0.006*2
0805Resistor	180	2	0.0208	£0.006
0805Resistor	1.6k	1	0.0208	£0.006
0805Resistor	2.2k	1	0.0208	£0.006
0805Resistor	2.7k	1	0.0208	£0.006
0805Resistor	3.6k	1	0.0208	£0.006
0805Resistor	7.5k	1	0.0208	£0.006
0805Resistor	24k	1	0.0208	£0.006
0805Resistor	30k	1	0.0208	£0.006

0805Capacitor	1.5n	2	0.152*2	£0.052*2
0805Capacitor	100n	2	0.08*2	£0.022*2
Electrolytic Capacitors	MCVVT016M1 51EA6L 150u	2	0.25*2	£0.075*2
Electrolytic Capacitors	MCVVT016M3 31FB3LSMD 330u	2	0.48*2	£0.14*2
Inductor	SRR4528A 150u	2	1.22*2	£0.2856*2
Rectifier	1N5819HW	2	0.25*2	£0.05*2
NPN BJT	BC817-40W	4	0.096*4	£0.04*4
ICs	MC33063AD	2	1.49	£0.6*2
Connector	M20-8770646	1	0.4	£0.28
battery	-	1	3.8	3.8
Total (Add shipping fees)	-	32	£11.2084	£6.8192
Total for 200 boards (Add shipping fees)	-	6400	-	£1363.84

- **PCB board**
The cost of manufacturing one 2-layer 5cm*5cm PCB is £25.0. But the cost of massive production (200 units) is £1.53 per PCB as the total area reaches $0.5m^2$. The total cost of manufacturing 200 PCBs is **£306.0**.
- **Assembly**
The assembling for one single board will take the technician (our team members) 0.5 hour which costs £10.
By comparison with that, automatic robots are used to pick and place the components, they are quicker, more accurate and cost less than the technicians. The robots we chose are provided by the outsourcing company. The price is £0.1 per component including the soldering tin used and materials loss, the total cost for 200 boards and 6400 components will be **£655** plus delivery fee.
- **Testing and Calibration**
The testing of all of the functionality of one single board following the verification procedure by a technician (our team members) took 1 hour which costs £20.
The technicians are asked to test and calibrate the boards. The salary of them is £20/h with tax and the approximation test time for a single board is 3 min following the manufacture test procedure on average. We cannot carry out the verification plan for every board, as this will take a long time and cost more. Most boards will undergo some basic tests and some of the boards will be randomly chosen for additional tests to ensure the quality. The details of the testing procedure are shown in the manufacturing test procedure.
The total cost for testing is **£70**.
- **Packaging**
The cost of packaging for one board by ourselves is £5.
The cost of packaging materials, including paper boxes or plastic bags, for 200 boards is £ 9.32 with tax.
The cost of packing by workers is around £12/h. It will take a total of 2 hours which includes carrying the boards from the manufacturing shop, sorting out the boards and packing following the users' order forms.
The total cost of packaging is **£33.32**.

Non-recurring cost:

The total non-recurring cost for manufacturing a single board or one unit from 200 boards is the same, which is.

- **Design**
This design took 10 hours and the salary of an engineer, including tax, pension and overheads such as cost of his or her desk and heating and lighting, is approximately £40/h.
The final cost is **£400**.
- **Making documents**

The time of producing the datasheet is approximately hours by engineers, including verifications, testing for data lists, plotting diagrams and making the documents. So, the total cost is **£120**.

- Tooling and support service

The cost of making the mechanical milling machine with 50mm*50mm is **£1890**.

The cost of programming the automatic robots is **£120**, within 3 hours of work by engineers.

The cost of support service requires 2 design engineers and 3 technicians to spend 20 minutes each day. The support took two days so the total cost of support service is **£73.33**.

The cost of equipment renting, including 1 DC power supply for 1 week **£38.50**, 1 Digital Multimeter for 1 week **£60.50**, 1 Oscilloscope for 1 week **£220**, 1 computers for designers for 1 week **£6.2** and license for 1 month use of Multisim **£9.99**. Totally **£335.19**.

*Assumption made:

We assume that we already have locations just like the teaching lab to produce the boards and we do not need to pay for the delivery to our customers. All technicians and engineers are well trained so the working time does not include the time for training. We also assume all the fabrication and testing can be done within one week.

Total cost:

For a single DPS: **£3059.7284**

For 200 DPSs: **£5366.68**

For one DPS from 200 DPSs: **£26.8334**

Reference:

Sample pricing for the materials purchased.

0805 resistors:

https://www.arrow.com/en/products/rc0805fr-0751rl/yageo?utm_currency=GBP&gad_source=1&gclid=Cj0KCQiAoeGuBhCBARIsAGfKY7wUM1RbuHDBuqldE3hFRMXxsHT1nttbcHwlhiBDW3DG8L7PEOWAUEUaAnYZEALw_wcB&gclid=aw.ds

0805Capacitor 1.5n:

[08055C152K4T2A KYOCERA AVX | Mouser United Kingdom](#)

0805Capacitor 100n:

[08055C104KAT2A KYOCERA AVX | Mouser United Kingdom](#)

MCVVT016M151EA6L 150u:

[MCVVT016M151EA6L Multicomp Pro, SMD Aluminium Electrolytic Capacitor, Radial Can - SMD, 150 µF | Farnell UK](#)

MCVVT016M331FB3LSMD 330u:

[MCVVT016M331FB3L Multicomp Pro, SMD Aluminium Electrolytic Capacitor, Radial Can - SMD, 330 µF | Farnell UK](#)

BC817-40W:

[BC817-40-QR Nexperia | Mouser United Kingdom](#)

1N5819HW:

[1N5819HW-7-F Multicomp Pro, DIODE, SCHOTTKY, 1A | Farnell UK](#)

SRR4528A-150uH:

[MCSDRH73B-151MHF Multicomp Pro, Power Inductor \(SMD\), 150 \$\mu\$ H, 690 mA | Farnell UK](#)

M20-8770646:

[M20-8770646 Harwin Inc. | Connectors, Interconnects | DigiKey](#)

MC33063AD:

<https://www.findchips.com/search/MC33063AD>

Packing bags:

https://www.polybags.co.uk/shop/100-percent-recycled-mailing-bag_p1655.htm?gad_source=1&qclid=Cj0KCQiAoeGuBhCBARIsAGfKY7z0tDZIWbbJk1z0dL9KFrzGFTFGq42KtcbSW_iwQGJN8AfRPVSbxcoaAnZwEALw_wcB

mechanical milling machine

<https://www.alibaba.com/showroom/mechanical-milling-machine.html>

Equipment renting:

<https://www.inlec.com/thurlby-thander-instruments-ts3021s-power-supply>

<https://www.inlec.com/metrix-mx51ex-digital-multimeter>

<https://www.inlec.com/agilent-technologies-hp-54520a-digitalising-oscilloscope>

<https://www.mygeex.co.uk/lease/hp-250-g9-15-6-intel-core-i7-16gb-512gb-ssd>

<https://www.multisim.com/pricing/>